

power supplies item table

No.	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
PS-1	AC input connector	Assembly	Receive external utility power into the power-supplies cabinet.	AC inlet connector, cable gland, grounding lug, strain relief.	Copper conductors, tin-plated copper terminals, steel or aluminum connector housing, polymer insulation.	This is the BINP 60 kV / 15 kW high-voltage source paper; it identifies the source inpre rectification. Page p.1. Quote: “AC 3- phase voltage.”	Material is standard electrical connector engineering; final material should be locked by connector vendor datasheet.
PS-2	Main disconnect switch	Assembly	Provide manual isolation of the incoming power for operation and service safety.	Main switch, actuator handle, terminals, mounting bracket, enclosure cutout.	Copper contacts, silver alloy contact surface, thermoset/thermoplastic body, steel mounting hardware.	This is the BINP high-voltage source paper; its design section saytains the input switch. Page p.2. Quote: “input switch.”	Material is switch-vendor specific; final material should be based on rated voltage/current and the lunar cabinet environment.
PS-3	EMI filter	Assembly	Reduce high- frequency noise conducted back to the power line.	Common-mode choke, differential- mode capacitors, line-to-ground capacitors, metal case, terminals.	Copper winding, ferrite or nanocrystalline core, film capacitors, aluminum or steel case.	This is the BINP high-voltage source paper; its Input Rectifier section states the lter. Page p.1. Quote: “eliminate high- frequency noise.”	This is the same BINP paper; it identilectromagnetic filter. Page p.1. Quote: “EMI filter F1.”
PS-4	Input rectifier	Assembly	Convert 3-phase AC input into DC voltage for the DC bus.	Three-phase diode bridge, rectifier terminals, mounting plate, thermal interface.	Silicon rectifier diodes, copper terminals, aluminum baseplate, ceramic isolation if module type.	This is the BINP high-voltage source paper; it dhe 380 V AC input to DC. Page p.1. Quote: “DC 550-600V.”	Material is rectifier-module specific; final material should be locked by rectifier supplier datasheet.
PS-5	DC-link capacitor bank	Assembly	Store and smooth the rectified DC bus energy before the inverter.	DC-link capacitors, bus connections, balancing resistors, mounting clamps.	Metallized polypropylene film capacitors or electrolytic capacitors, copper busbar, polymer/metal capacitor case.	This is the BINP high-voltage source paper; it idenfier filter capacitors. Page p.1. Quote: “filter C1-C4.”	This is a KEMET white paper on industrial power film capacitors; it supp(mouser.it)for power capacitor use. Quote: “Polypropylene film.”
PS-6	Ripple damping inductor	Assembly	Reduce current ripple in the rectified DC input stage.	Inductor winding, magnetic core, bobbin, terminals, mounting bracket.	Copper winding, ferrite/iron/nanocrystalline core, polyimide/Nomex insulation, steel or aluminum bracket.	This is the BINP high-voltage source paper;a rectifier filter element. Page p.1. Quote: “L1 C1-C4.”	Material is inductor-vendor specific; final material should be locked by magnetic-component supplier datasheet.
PS-7	Full-bridge inverter	Assembly	Convert the DC bus into high-frequency AC to drive the transformer chain.	IGBT bridge, anti-parallel diodes, gate- driver interface, snubber capacitors, laminated busbar, heatsink interface.	IGBT module, AlN or alumina DCB substrate, copper or AlSiC baseplate, copper busbar, aluminum wire bonds.	This is the BINP high-voltage source paper; it defin and its frequency. Page p.1. Quote: “20 kHz frequency.”	This is an IGBT module reliability applicati(rellpower.com)al module construction. Quote: “AlN ... alumina DCB.”
PS-8	Matching circuit	Assembly	Reduce transients, compensate transformer leakage/parasitics, and improve efficiency.	Matching capacitor, two matching inductors, high- current terminals, mounting bracket, thermal path.	Copper winding, ferrite/nanocrystalline/amorphous core, polypropylene film capacitor, ceramic standoff.	This is the BINP high-voltage source p C5, L2, and L3 are used. Page p.1. Quote: “C5, L2, L3.”	This is the KEMET power film capacitor white paper; it supports pol(mouser.it)er-capacitor dielectric. Quote: “Polypropylene film.”

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PS-9	Primary isolation transformer	Assembly	Provide galvanic isolation and couple the inverter output into the HV transformer input chain.	Primary winding, secondary winding, magnetic core, insulation system, terminals, mounting frame.	Copper winding, ferrite or nanocrystalline core, polyimide/Nomex insulation, epoxy or varnish if not in vacuum.	This is the BINP high-voltage source paper; it lists the isolation tranter design. Page p.1. Quote: "isolation transformer T1."	Material choice is engineering selection; final lock requires transformer vendor specification.
PS-10	Control-electrode bias supply	Assembly	Bias the control electrode to regulate beam current and support beam modulation.	Isolated low-current HV supply, modulator switch, output filter, current monitor, voltage monitor, HV output connector.	HV MOSFET/IGBT switch, ceramic standoffs, HV resistors, film capacitors, shielded HV wire.	This is the BINP 60 keV / 30 kW electron-beam facility paper; it states thates the control electrode. Page p.3. Quote: "0 to 600 V."	This is TT Electronics' high-voltage resistor application note; it supports high-voltage(ttelelectronics.com)vider/bias circuits. Quote: "high voltage resistors."
PS-11	Cathode heater supply	Assembly	Provide stabilized high current to heat the thermionic cathode.	Isolated current source, high-current busbar, current sensor, output terminal, thermal protection, cable interface to gun.	Copper busbar, high-current MOSFET/IGBT module, ceramic terminal block, manganin or Hall current sensor.	This is the BINP electron-beam facility paper; it states that the h cathode heating current. Page p.3. Quote: "0 to 100 A."	This is the IGBT module application note; it suppo(rellpower.com) power-module construction. Quote: "copper bus-bars."
PS-12	Auxiliary low-voltage DC supply	Assembly	Provide common low-voltage DC power for sensors, controls interfaces, motors, valves, coil drivers, wire feeder electronics, and positioning electronics, excluding the cathode heater supply and control-electrode bias supply.	AC/DC or DC/DC module, low-voltage DC bus, output terminals, branch fuses, status output, remote inhibit input.	Copper wire, tin-plated terminals, FR-4 PCB, aluminum enclosure, polymer insulation, ferrite EMI choke.	This is the NASA EBF ³ patent; it defines the power distribution subsyiple subsystems. Page p.12. Quote: "distributing power."	Material choice is standard industrial power-supply engineering; final materials require vendor datasheet.
PS-13	Low-voltage distribution panel	Assembly	Split auxiliary DC power into protected branches for each load group.	Terminal blocks, branch fuses, circuit breakers, relays, labels, harness connectors.	Copper busbar, tin-plated terminals, steel DIN rail, polyamide terminal blocks, aluminum/steel panel.	This is the NASA EBF ³ patent; it says the power distribution sub protects power. Page p.12. Quote: "apportioning power."	Material is electrical-panel engineering; final lock requires terminal-block, fuse, and breaker datasheets.
PS-14	Multi-channel driver module	Assembly	Provide controlled electrical outputs for magnetic lens coils, trajectory correctors, deflection/scanning coils, wire-feeder motor, and four-axis positioning motors.	Driver channels, MOSFET/IGBT power stages, current sensors, output filters, motor/coil connectors, branch fuses, protection circuits, interface to controls.	Power MOSFET/IGBT modules, copper busbar, film capacitors, ferrite inductors, aluminum heatsink, shielded motor/coil cable.	60keV 30kW electron beam- BINP Novosibirsk Page p.1 / p.3; Quote: "deflect a beam," "magnetic lens and correctors currents." NASA patent, p.10 / p.13; Quote: "wire feed motor," "positioning Subsystem motor."	Material selection is power-electronics engineering; final parts require driver, MOSFET/IGBT, cable, and connector supplier datasheets.
PS-15	Power-electronics thermal management hardware	Assembly	Remove heat from IGBTs, rectifiers, inductors, and DC-link components.	Heatsink, cold plate, thermal interface material, coolant fittings, conduction straps, temperature sensors.	Aluminum heatsink, copper cold plate, copper tubing, thermal grease/pad, stainless fittings.	This is the BINP high-voltage source paper; it states that distilled water cd other elements. Page p.2. Quote: "cool IGBT switches."	This is ATS/QATS product information for liquid cold plates used to cool IGB(Qats) smiconductors. Quote: "liquid cooling is required."